Project Synopsis

Programme: MBA (II Year) — Project Study (4 Credits)
Category: (c) Business Plan for an Entrepreneurial Venture

Student: ____________________ **Roll No.:** ____________________
Guide: ____________________

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1. Title

Business Plan for "ASDLC": Commercializing an Open-Source Governance Tool for AI-Assisted Software Development

2. Background and Rationale

Software development is undergoing a structural shift: AI coding agents (Claude Code, OpenAI Codex, Cursor, DeepSeek-based tools) now write a large share of production code. Industry analysts project enterprise adoption of agentic AI to grow from under 1% (2024) to roughly 33% by 2028 (Gartner).

This speed has created a new, largely unaddressed management problem: **concept drift**. Because each AI session works without memory of previous sessions, business rules get implemented inconsistently across a codebase — duplicated logic, contradictory definitions, half-completed changes — causing recurring defects and rising maintenance cost. Research indicates AI-agent usage measurably increases code-analysis warnings (~30%) and code complexity (~41%). In management terms, this is an organizational-memory and quality-governance failure, not merely a technical one.

ASDLC is an early-stage open-source venture (founded by a practitioner; the technology already exists as a working command-line tool with pilot evidence from a production application) that addresses this gap with a governance layer: a human-owned registry of business-rule definitions, automated quality gates, and a continuous audit loop. Established competitors (GitHub Spec Kit, BMAD-METHOD, Amazon Kiro) focus on structuring *new* AI work; none owns the *drift-governance* niche. The venture requires what it currently lacks — a rigorous business plan — which is the contribution of this project.

Role clarity: *the software was developed by the venture's founder. This project contributes the business analysis: market research, competitive strategy, business model design, and financial planning.*

3. Problem Statement

Organizations adopting AI-assisted development lack mechanisms to keep AI-generated code consistent with business intent, leading to recurring defects and eroding trust in AI development. Can a commercially sustainable venture be built around an open-source governance tool that addresses this problem, and what business model, pricing, and go-to-market strategy would make it viable?

4. Objectives

1. To assess the market opportunity for AI-code-governance tooling (market size, growth, segmentation of potential adopters).
2. To analyze the competitive landscape of AI development workflow tools and identify a defensible positioning for the venture.
3. To conduct primary research among software-team decision-makers on AI-coding quality pain points and willingness to pay for governance solutions.
4. To design a business model (open-core: free CLI + paid hosted services) with pricing strategy and revenue projections.
5. To prepare a complete, investor-ready business plan including go-to-market plan, operations plan, risk analysis, and 3-year financial projections.

5. Literature and Secondary Sources (indicative)

- **Software quality and duplication:** Hunt & Thomas, *The Pragmatic Programmer* (DRY principle, 1999); Fowler, *Refactoring* (code smells, “shotgun surgery”); Cunningham (1992) on technical debt; Conway (1968) on organizational mirroring in systems.
- **AI-assisted development:** recent studies on agentic coding quality effects (arXiv, 2026); industry analyses of spec-driven development frameworks (GitHub Spec Kit, BMAD-METHOD, Kiro) and agent harnesses.
- **Business models:** literature and cases on open-source commercialization (open-core, support/services models — GitLab, HashiCorp, Docker).
- **Adoption theory:** Rogers’ diffusion of innovations; technology acceptance in developer-tools markets.

6. Research Methodology

- **Research design:** descriptive + exploratory, culminating in a business plan (applied output).

- **Primary data:** (a) structured survey of 30–50 software professionals (CTOs, engineering managers, senior developers) on AI-coding adoption, drift pain, current mitigation, and willingness to pay; (b) 5–8 semi-structured interviews with engineering leaders; (c) founder interviews for product and cost inputs.
- **Secondary data:** analyst reports, competitor documentation and pricing, open-source adoption metrics (GitHub stars/downloads as proxies), published research.
- **Pilot evidence (secondary, with permission):** anonymized results from the tool's pilot on a production insurance CRM (defect-recurrence causes; gate vs. registry detection rates) used as product-validation evidence.
- **Analysis:** market sizing (TAM/SAM/SOM), Porter's Five Forces, SWOT, competitor feature/pricing mapping, survey statistics, break-even and scenario-based financial projections.

7. Scope and Limitations

Scope: global developer-tools market with focus on organizations adopting AI coding agents; business-plan horizon of three years. **Limitations:** the venture is pre-revenue, so financials are projection-based; pilot evidence derives from a single anonymized organization; fast-moving market conditions may date specific competitor details.

8. Expected Outcomes / Chapterization

1. Introduction and industry background
2. Literature review
3. Research methodology
4. Market analysis and primary research findings
5. Competitive analysis and strategic positioning
6. The business plan: product, model, pricing, go-to-market, operations, team, risks
7. Financial projections and funding requirements
8. Conclusions and recommendations

9. References

(To be formatted per programme guidelines; indicative sources listed in §5.)